STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 1: Units and Measurements

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Seven.

Q2 - 1 mark

kelvin (K).

Q3 - 1 mark

Three.

Q4 - 1 mark

[LT^-1].

Q5 - 1 mark

[MLT^-2].

Q6 - 1 mark

Yes.

Q7 - 1 mark

No.

Q8 - 1 mark

pascal (Pa).

Q9 - 2 marks

12.5.

Q10 - 2 marks

[MLT^-1].

Q11 - 2 marks

ut and at^2 both have dimension L; homogeneous.

Q12 - 2 marks

20 m s^-1.

Q13 - 2 marks

Four.

Q14 - 2 marks

Both are [ML^2T^-2].

Q15 - 3 marks

Equating dimensions gives a=, b=-, so T proportional to sqrt(l/g).

Q16 - 3 marks

4pir^3/3=33.5 cm^3.

Q17 - 3 marks

[M^-1L^3T^-2].

Q18 - 3 marks

51.68 cm^2 rounds to 52 cm^2.

Q19 - 4 marks

a=, b=-; T proportional to sqrt(m/k).

Q20 - 4 marks

18.2 cm^3; extra digits are unsupported.